

Tobacco Consumption Habits in Argentina: Causal Evidence from a New Regulation

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Abstract

Modifiable risky behaviors such as smoking tobacco are a major determinant of premature death in developed and developing countries. In this paper, I study the effects of non-price interventions on smoking habits in the context of Argentina's 2011 anti-smoking law. I interact previous state-level legislation with the national law to identify the effect of the policy on smoking prevalence and average daily cigarette consumption. I find that the probability of being a current smoker decreased by 2.7 percentage points, or 8.7% of the baseline mean, with a corresponding increase in the probability of being a never smoker. This reduction in smoking prevalence is driven in part by light smokers exiting the smoking pool, generating a rightward shift in the distribution of daily cigarette consumption among continuing smokers. These changes are accompanied by a reduction in COPD hospitalization rates of approximately 10% relative to the pre-policy mean, with no detectable effect on lung cancer hospitalizations over the same horizon.

Keywords: risky behaviors, tobacco control legislation, event study, developing countries, health outcomes

JEL codes: I12, I18, K32, D12

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I Introduction

Modifiable risky behaviors, such as smoking tobacco, are a major determinant of premature death in both developed and developing countries (Blecher, 2008; Cawley and Ruhm, 2011). The World Health Organization estimates that tobacco use is the leading risk factor in high-income countries, accounting for 18% of deaths, while in middle-income countries—where 80% of smokers worldwide live—it is the second most important risk factor, responsible for about 11% of deaths (WHO, 2009). Moreover, Goodchild, Nargis and Tursan d’Espaignet (2018) estimate that 40% of the health care expenditure attributable to smoking-related diseases occurs in developing countries, underscoring the substantial burden these countries face.

Although tobacco has long been heavily taxed, non-price policies have become increasingly common over the last two decades (DeCicca, Kenkel and Lovenheim, 2022). The most frequently implemented are place-based bans and graphic warning labels: by 2025, at least 140 countries and territories required graphic warnings to be printed on cigarette packages.¹ Such policies are particularly attractive in low- and middle-income countries, where tobacco use is concentrated among low-income households (Crawford and Le Nestour, 2019; Fuchs, Gonzalez Icaza and Paz, 2019). The addiction model of Bernheim and Rangel (2004) provides a natural lens to understand why these instruments can change behavior absent any change in price. Their framework places a high value on “cognitive” policies that improve opportunities for self-regulation—the suppression of environmental cues, for example through limits on advertising and on where smoking is permitted, and the dissemination of counter-cues, such as graphic warnings (pictorial images of smoking’s health harms printed on cigarette packages). Place-based bans force smokers outdoors and raise the time and discomfort cost of smoking, while warning labels raise the perceived long-term cost of tobacco use. Argentina’s 2011 law bundles both instruments at a national scale, offering a rare opportunity to study cognitive tobacco policy in a middle-income country.

In this paper, I examine the effect of these non-price interventions on smoking and health outcomes in a middle-income country, Argentina, where smoking is responsible for 13.2% of deaths and a total societal burden estimated at 1.2% of the national GDP

¹In 2001, Canada became the first country in the world to introduce graphic warnings on cigarette packages, and was quickly followed by many others (Azagba and Sharaf, 2012). In the United States, the 2009 Family Smoking Prevention and Tobacco Control Act required graphic warnings on cigarette packages, but implementation has been delayed for over a decade by litigation from cigarette manufacturers and, as of 2026, the rule has yet to take effect.

(Alcaraz et al., 2016; Palacios et al., 2023). In 2011, the Argentinean government enacted an anti-smoking law that banned smoking in public spaces, including bars, restaurants, and public and private workplaces; restricted cigarette sales; and mandated the inclusion of graphic tobacco warnings on cigarette packages. The Argentinean case offers a compelling setting for two reasons. First, the policy was not accompanied by any change in taxation or cigarette prices, so it represents a *pure* non-price intervention. Second, Argentina is a federal country, like the United States, in which states retain autonomy to regulate smoking and access to tobacco products even though taxation is exclusive to the national government. Before 2011, no state had legislation as strict as the new national law, and the degree of tightening varied with the stringency of each state's prior regulation. I exploit these state-level differences as a source of exogenous variation in exposure to the national policy: states with more lenient pre-existing rules (treated states) were more exposed to the new regulation than states with stricter rules (comparison states).

Data for this paper come from three main sources. First, to estimate the effects on smoking, I use individual-level data from two national surveys for three pre-policy years (2005, 2008, and 2009) and three post-policy years (2013, 2017, and 2018), which include geographic location, demographic characteristics, and cigarette consumption.² Second, to study health outcomes, I use restricted-access administrative data on hospital discharges provided by the Argentinean National Center for Health Statistics, which I aggregate at the state level. Third, I compile data on tobacco regulation from state statutes and laws, coding a total of 47 state-level regulations to characterize the regulatory environment of each state before the 2011 national law was enacted. My identification strategy compares the smoking behavior of individuals in lenient states with that of individuals in strict states. I show that treated and comparison states were on similar pre-policy trends and that individuals in the two groups were balanced in observable characteristics before 2011. I further show that pre-2011 treatment status is unrelated to political affiliation or tobacco-producer status, mitigating concerns that the decision to regulate tobacco early was driven by political considerations or by the economic interests of the tobacco industry.

I find that the anti-tobacco regulation effectively reduced smoking participation. By 2013, the first post-policy survey year, the probability of being a current smoker fell by approximately 2.7 percentage points—8.7% of the baseline mean—with a correspond-

²The two surveys are the National Survey on Risk Factors (ENFR) and the National Survey on Consumption Prevalence of Psychoactive Substances (EnPreCoSP).

ing increase of about 2 percentage points (3.7% of the baseline mean) in the probability of being a never smoker. Both effects are statistically significant and consistent with the law operating simultaneously through deterrence among potential smokers and cessation among existing ones. The point estimates remain similar in magnitude when the sample is extended to 2017 and 2018, suggesting that the effects are persistent; however, the long-run estimates are less precise, reflecting the smaller number of post-policy observations, and are not statistically distinguishable from zero.

The reduction in smoking prevalence is driven in part by the exit of light smokers from the smoking pool. Examining the distribution of daily cigarette consumption among continuing smokers, I find a rightward shift: by 2013, the share of smokers consuming between zero and five cigarettes per day fell by 10.73 percentage points, while the share consuming more than 20 cigarettes per day rose by 2.69 percentage points. This compositional shift carries an important policy implication—the 2011 law appears to have reached a relatively receptive margin of the smoking population first, so that further reductions in prevalence will likely require complementary interventions targeted at heavier, more addicted smokers. The effects are also heterogeneous across the population. The largest reductions in current smoking are concentrated among the youngest adults (18–25), who respond primarily by not initiating, and among adults aged 45–55, who respond through cessation; I find no effect for adults aged 55–65, consistent with the greater difficulty of quitting after a longer history of nicotine addiction. Across education groups, the absolute reduction in smoking is broadly similar, but in relative terms it is larger for college graduates.

The impacts on smoking are reflected in slightly better health outcomes. Using hospital-discharge data, I find suggestive evidence of a decline in chronic obstructive pulmonary disease (COPD) hospitalizations, with point estimates around 10% of the pre-policy mean that reach statistical significance in 2013 and 2014. I find no detectable effect on lung cancer hospitalizations over the same horizon, consistent with the longer time required to reverse the onset of that disease. Because the discharge data are not available beyond 2014, I cannot determine whether these health effects persist over a longer horizon; I therefore interpret them as suggestive evidence of an improvement in population health associated with stricter tobacco regulation.

This paper contributes to two strands of the literature. First, it speaks to a large literature in health economics evaluating place-based smoking bans and other non-price regulations, where the evidence remains mixed. Research in North America finds mixed impacts of smoking bans on smoking behavior ([Adda and Cornaglia, 2010](#);

Bitler, Carpenter and Zavodny, 2010; Burton, 2025; Carpenter, Postolek and Warman, 2011), while evidence from Europe points to systematic reductions of small magnitude (Buonanno and Ranzani, 2013; Pieroni et al., 2013; Sureda et al., 2014). As DeCicca, Kenkel and Lovenheim (2022) note in their review, the evidence from even the best-designed studies on how these policies affect smoking behavior is less than definitive. The effectiveness of non-price restrictions in low- and middle-income countries—where the vast majority of smokers live—has been studied much less, and the existing work relies largely on cross-country (Abascal et al., 2012; Blecher, 2008) or before-and-after (e.g., Thrasher et al., 2010) comparisons. I contribute credible causal estimates of the effect of a pure non-price policy bundle in a middle-income setting, using a difference-in-differences design that exploits within-country variation in pre-existing regulation and allows direct tests of parallel pre-trends.³ This evidence complements the findings of Harris, Balsa and Triunfo (2015), who exploit the staggered rollout of Uruguay’s campaign to show that non-price regulation of packaging and marketing accounted for most of the rise in smoking cessation among pregnant women.

Second, this paper advances the economic research on graphic warnings and on the downstream health effects of tobacco regulation. Recent economics evidence on graphic warnings comes primarily from experimental settings: in a stated-preference discrete-choice experiment, Kenkel et al. (2023) find that graphic labels lower the share of respondents choosing cigarettes and raise quitting, driven more by fear and disgust than by improved health knowledge. Evidence on how such warnings affect realized behavior under an actual national policy remains scarce. I provide such evidence, and I push further by asking whether the resulting reductions in smoking translate into measurable health gains.⁴ The central empirical challenge is that many smoking-related conditions—heart disease, lung cancer, premature mortality—develop over long horizons, making it difficult to identify the lag between policy and disease onset (DeCicca, Kenkel and Lovenheim, 2022). I address this by exploiting administrative data on hospital discharges rather than mortality: the lag to observable effects is shorter, and hospitalization data capture health gains from cessation that need not appear in mortality.

³My baseline classifies states as treated (lenient) or comparison (strict) according to the stringency of their pre-existing regulation. Callaway, Goodman-Bacon and Sant’Anna (2024) show that, in difference-in-differences designs with a continuous treatment, a binary treatment indicator recovers a well-defined average treatment effect for treated units under a parallel-trends assumption analogous to the binary case.

⁴Harris, Balsa and Triunfo (2015) take this step for Uruguay by linking cessation to infant birth weight; I instead study adult chronic-disease hospitalizations (COPD and lung cancer), an outcome closer to the long-run harms that motivate tobacco regulation.

For example, in the case of COPD, cessation and reduced exposure to cigarette smoke slow the excess decline in lung function (Eklund et al., 2012), a mechanism consistent with the relatively fast gains I find.

Beyond its specific contributions, this paper speaks to the broader stakes of tobacco-control policy. The health consequences of smoking unfold over decades, so the gains from reducing smoking today accrue largely in the future: exploiting variation in cigarette taxes, Friedson et al. (2023) show that lower smoking translates into measurable improvements in longevity and health in the long run. The distribution of those gains matters as much as their size. I find that the 2011 law reduced smoking proportionally more among the more educated, and recent evidence identifies precisely this kind of differential decline in smoking as a central driver of the widening gap in midlife mortality between college- and non-college-educated Americans (Foote et al., 2025). Policies that lower smoking unevenly across the population can therefore narrow or widen health inequalities, depending on whom they reach—an important consideration for the design of non-price tobacco regulation in middle-income countries, where smoking is increasingly concentrated among the poor.

The rest of the paper is organized as follows. Section II presents background on non-price tobacco regulation and the Argentinean national law. Section III describes the data. Section IV presents the empirical strategy and provides evidence supporting the identification assumption. Section V examines the effects of the national regulation on smoking, along the extensive and intensive margins and across population groups. Section VI presents the effects on health outcomes. Section VII reports robustness checks, and Section VIII concludes.

II Non-price Regulation and the Argentinean National Law

The Bernheim and Rangel (2004) model of addiction places a high value on policies that improve opportunities for self-regulation and identifies a central role for “cognitive” policies, including the suppression of certain environmental cues (e.g., through limitations on advertising) and the dissemination of counter-cues (e.g., graphic warnings). Restrictions on smoking in public places and private worksites impose additional costs on smokers by forcing them outdoors, raising the time and discomfort associated with smoking. Similarly, new information on the health consequences of tobacco use can raise the perceived long-term costs of smoking.

Argentina’s 2011 policy can be analyzed through the lens of this model. By ban-

ning smoking in public spaces such as restaurants and bars, the law imposes direct costs on smokers and limits their exposure to environmental cues. By mandating the incorporation of Tobacco Graphic Warnings (TGW), it disseminates counter-cues at the point of purchase. These warnings consist of short messages about smoking’s health consequences, accompanied by a graphic image on one side of the package (see Figure A.1). Images are designed by the National Department of Health and updated every eighteen months to attenuate potential wear-out effects. After the law was approved, producers had up to six months to incorporate the new designs on their packages.

Argentina is a federal country and, as in the United States, states retain some autonomy to implement their own regulations. While taxation of tobacco products is exclusive to the national government, each state can regulate smoking and access to tobacco products. Before 2011, no state had legislation as strict as the new national law. Crucially, the degree of tightening varied with the stringency of each state’s prior regulation, generating the regional variation I use to identify the causal effect of the national policy.

To operationalize this variation, I collected legislation data for Argentina’s 23 states and the capital city. States with strict pre-existing regulation serve as the control group, while those with weaker prior regulation constitute the treatment group. Table A1 presents a summary of the state laws, including a short description and the treatment status assigned to each state.⁵ Figure 1 illustrates the geographic distribution of treatment status across states. Importantly, the 2011 law bundled two distinct non-price interventions — public smoking bans and graphic warning labels — which cannot be separately identified in the empirical analysis that follows. However, both components operate through the same theoretical channels in the Bernheim and Rangel (2004) framework: public smoking bans suppress environmental cues by removing the social context in which smoking occurs, while graphic warnings disseminate counter-cues at the point of purchase. The estimates presented below are therefore best interpreted as the effect of a realistic policy package rather than of any single instrument, reflecting the way comprehensive tobacco control legislation is typically designed and implemented in practice.

⁵Section B provides details on the criteria and definitions used in coding the state laws.

III Data and Descriptive Statistics

This section describes the data sources leveraged in the empirical analysis. I combine data from two nationwide surveys and hospital discharge records to construct outcome variables.

Data on consumption of tobacco come from two nationwide surveys: the National Survey on Risk Factors (ENFR) and the National Survey on Prevalence of Consumption of Psychoactive Substances (EnPreCoSP). These surveys provide cross-sectional data with state-level coverage for 2005, 2008, 2009, 2011, 2013, 2017, and 2018. I observe individual consumption of tobacco products and individual characteristics, including age, educational attainment, marital status, employment status, and household income quintiles.⁶ Unfortunately, there is a considerable reduction in sample size in 2017 and 2018 relative to prior years which contributes to the wider confidence intervals of the long-run estimates.

I consider an individual to be a current smoker if she has ever smoked more than 100 cigarettes or if at the moment of the survey she smokes every day or some days. Individuals who have smoked a few times in their life and for whom smoking has not turned into a daily habit are considered never smokers. This definition is close to the one used by [Carpenter, Postolek and Warman \(2011\)](#) with Canadian data, who defined current smokers as those reporting smoking daily or occasionally.⁷

The consumption of tobacco is self-reported, which raises the concern of under-reporting; in that case, my estimates can be interpreted as a lower bound. I leverage data on sales to construct an additional estimate of consumption and compare the evolution of aggregate trends. Table [A2](#) presents comparisons of aggregate consumption from sales and survey data. Survey data mirror the main trends in consumption shown in sales data, and under-reporting appears stable across years: survey data captures about 58% of consumption relative to sales data.

Data on health outcomes come from the Argentinean National Center for Health

⁶During the period under study, the Argentinean economy experienced high inflation, which challenges the use of nominal income due to the lack of reliability on official estimates of the Consumer Price Index (for a discussion see, [Cavallo, 2013](#)). I use household income quintiles, computed using the distribution of income constructed each year from household employment surveys, as a proxy for individual purchasing power.

⁷One potential limitation is that *social smokers* might not be classified as smokers, yet the legislation change could affect their consumption since restaurants and bars are where these individuals are most likely to smoke. This is a conservative definition that may lead to underestimating the effects of the policy on the share of current smokers.

Statistics.⁸ Upon a patient’s visit to a hospital, the admissions area must fill a *Statistical Hospitalization Report* with the patient’s basic demographic information. Once the patient is examined, the report is updated with the main diagnosis, secondary diagnosis, surgeries and other medical interventions, and type of discharge. Diagnoses are classified following the International Statistical Classification of Diseases and Related Health Problems, tenth revision. I gained restricted access to these administrative discharge data, which cover the period 2008–2014. Each observation corresponds to one hospitalization episode, for which I observe state identifiers, individual characteristics (gender and age), and the main diagnosis. I collapse these data to compute hospitalization rates for two diagnoses of interest: Chronic Obstructive Pulmonary Disease (COPD) and respiratory system related cancer.⁹ These data do not include individual identifiers or smoker status; nonetheless, they allow me to capture changes in population health before and after the national law.

Table 1 shows mean outcomes for the pre-policy period. Strict and lenient states have similar shares of never and current smokers, and the equality of means tests confirm this balance. The distribution of cigarettes smoked per day, however, shows some differences: strict states have more mass at lower consumption levels (0–5 cigarettes per day) while lenient states show slightly more mass at higher levels (20+). Turning to health outcomes, lung cancer prevalence rates are similar across groups. COPD rates differ somewhat—0.34 versus 0.49 per 1,000 in lenient and strict states—but the gap is small in absolute terms and not economically meaningful. These baseline differences reflect variation in underlying health conditions rather than systematic differences in smoking behavior.

IV Empirical Strategy

My identification strategy relies on state-level differences in the strength of tobacco regulations before the national law was passed. Figure 1 depicts the geographic distribution of treatment status: lenient states, where the national law implied a tightening in regulation, constitute the treated group, while strict states serve as the comparison group and provide the counterfactual smoking behavior. I estimate the causal effect of the national regulation via the following regression, run over samples of individuals

⁸In Spanish, Direccion de Estadisticas e Informacion de Salud (DEIS). This agency is analogous to the U.S. National Center for Health Statistics.

⁹COPD is defined using ICD 10th revision codes J41, J42, J43, and J44. Respiratory system related cancer is defined using ICD 10th revision codes C30, C33, and C34.

in treated and comparison states observed in three pre-policy years (2005, 2008, and 2009) and three post-policy years (2013, 2017, and 2018):

$$y_i = \sum_{t \neq 2011} \delta_t [Treat_s \cdot (\text{year} = t)] + \beta' X_{is} + \Gamma' X_{st} + \alpha_s + \alpha_t + \varepsilon_i, \quad (1)$$

where s indexes the state where individual i is observed in year t and y_i is the outcome of interest. $Treat_s$ indicates whether state s serves as a comparison state ($Treat_s = 0$) or as a treated state ($Treat_s = 1$). Because the survey waves are unevenly spaced, I interact $Treat_s$ with a full set of survey-year indicators rather than a continuous time-since-treatment variable, omitting the 2011 wave—the year the law was passed—as the reference category. Each δ_t measures the difference in outcomes between treated and comparison states in wave t relative to 2011. The pre-policy coefficients (2005, 2008, 2009) capture whether treated and comparison states were on similar trends before the policy change. The coefficients for the post-policy periods capture the effect of the national law two, six, and seven years after the law was passed, respectively. X_{is} is a vector of individual-level control variables including age, gender, educational attainment, and income. I also include state-time varying controls: private employment and population. State fixed effects (α_s) control for variation in outcomes across states that is constant over time. Time fixed effects (α_t) control for variation in outcomes over time that is common across all states. The term ε_i is an individual error term. Standard errors are block-bootstrapped at the state level with 300 replications.¹⁰ The event-study figures present point estimates and symmetric percentile-t confidence intervals. These are constructed by scaling the bootstrap standard deviation of $\hat{\delta}_\tau$ by the $(1 - \alpha/2)$ percentile of the bootstrap t-statistics and adding and subtracting the result from the point estimate.¹¹

My baseline specification uses a binary treatment indicator, classifying states as treated (lenient) or comparison (strict) based on their pre-existing regulatory strin-

¹⁰Using cluster-robust standard errors permits both error heteroskedasticity and flexible error correlation within clusters. With a small number of clusters, cluster-robust standard errors are downward biased (Cameron, Gelbach and Miller, 2008); block-bootstrap procedures yield more accurate inference in this setting. Block-bootstrap standard errors maintain the autocorrelation structure within groups by keeping observations that belong to the same group together in a “block,” sampling groups rather than individual observations.

¹¹To see this, define $\hat{\delta}_\tau^{(b)}$ as the estimate of δ_τ from the bootstrap sample b . Then, the estimated confidence interval with confidence level α is: $\widehat{CI}_{boot} = [\hat{\delta}_\tau - \text{sd}(\hat{\delta}_\tau^{(b)}) \times q^*(1 - \alpha/2), \hat{\delta}_\tau + \text{sd}(\hat{\delta}_\tau^{(b)}) \times q^*(1 - \alpha/2)]$, where $q^*(1 - \alpha/2)$ is estimated from a bootstrap simulation by sorting the bootstrap t-statistics and taking the $(1 - \alpha)$ percentile. Symmetric percentile-t confidence intervals have the advantage of being centered in the point estimate of the coefficient of interest.

gency. Callaway, Goodman-Bacon and Sant’Anna (2024) show that, in differences-in-differences designs with continuous treatments, a binary treatment dummy can recover a well-defined average level treatment effect for treated units under a parallel trends assumption analogous to the binary case. Thus, my main assumption is that comparison states would have followed the same trend as lenient states in the absence of the national legislation. The specification in equation 1 allows me to test this assumption directly: for each outcome of interest, I estimate pre-period coefficients that capture whether treated and comparison states were on similar trends prior to the policy change. The remainder of this section provides additional evidence in support of this assumption: I examine balance in individual and state-level characteristics in the pre-policy period.

IV.A Balance in Individual Pre-Policy Characteristics

I test whether individuals in treated and comparison states were statistically indistinguishable in terms of their observed characteristics in the pre-policy period.¹² Balancing test results are presented in Table 2. The estimated differences indicate that individuals are statistically indistinguishable in terms of age, gender, marital status, and employment status; while some of these differences are statistically significant, they are not economically meaningful. I find evidence that individuals in strict and lenient states differ in terms of education and income. Strict states have slightly lower shares of college-educated individuals relative to lenient states. In terms of income, strict states have more mass in the fourth quintile while lenient states have more mass in the first and second quintiles. To account for these differences, I include controls for household income, age, gender, employment status, and educational attainment in every specification.

IV.B Selection into Strict Regulation

For the comparison between strict and lenient states to be valid, both groups need to be comparable in terms of observable and unobservable characteristics. To shed light on this, I examine what predicts whether a state enacted strict tobacco regulation before 2011. I estimate a cross-sectional regression at the state level in which the dependent variable is an indicator for having strict pre-existing regulation, and the regressors

¹²I present pooled results only. Conclusions on balance on observable characteristics do not change when the analysis is done by year.

include the political affiliation of the state government, an indicator for whether the state is a tobacco producer, measures of tobacco consumption prevalence and COPD hospitalization rates, and demographic and economic characteristics. Table 3 shows these results.

Neither political affiliation nor tobacco producer status predict the likelihood of enacting strict regulation, which is reassuring as it suggests that treatment status was not driven by political considerations or economic interests in the tobacco industry. Similarly, economic characteristics such as employment and unemployment rates do not predict treatment status, suggesting that the decision to regulate tobacco prior to 2011 was not shaped by the economic conditions of the state. States with higher COPD prevalence and a larger share of young population were more likely to have enacted strict regulation prior to 2011, consistent with health-conscious and demographically younger states taking a more proactive stance on tobacco control. Overall, the evidence suggests that treatment status is not systematically related to political or economic factors, lending support to the identifying assumption.

V Effects of the 2011 National Law on Tobacco Consumption

In this section, I examine the effects of the bans on consumption and the implementation of TGW on (i) extensive margin outcomes, (ii) intensive margin outcomes, and (iii) heterogeneous effects.

V.A Extensive Margin Outcomes

Smoking prevalence can fall through two distinct margins: individuals who would have started smoking decide not to (the never-smoker margin), and individuals who already smoke decide to quit (the former-smoker margin). The 2011 law plausibly operates through both channels simultaneously. To assess the relative importance of these two channels, Figure 2 presents estimates of the causal effect of the national law on these two margins.

Comparison and treated states were on similar trajectories in smoking outcomes prior to the passage of the national law. The estimates for 2005, 2008, and 2009 are small in magnitude and statistically indistinguishable from zero. Turning to the post-policy estimates, the short-run results—identified using data through 2013—reveal a clear effect of the national law on both margins. The probability of being a never smoker increased by approximately 2 percentage points by 2013 (or 3.7% of the baseline mean),

while the probability of being a current smoker fell by roughly 2.7 percentage points (or 8.7% of the baseline mean), both statistically significant. These short-run effects are consistent with the law operating quickly through both the initiation and cessation channels: the imposition of public smoking bans and graphic warning labels raised the cost of smoking inducing both deterrence among potential smokers and cessation among existing ones.

The long-run estimates, which extend the sample through 2017 and 2018, suggest that these effects are persistent. The point estimates remain similar in magnitude to the short-run estimates on both margins, indicating no meaningful attenuation of the policy's impact over time. However, the confidence intervals widen considerably at longer horizons, likely reflecting the smaller number of post-policy observations available in the long-run sample. As a result, the long-run estimates are not statistically distinguishable from zero, which is consistent with a loss of statistical power driven by smaller samples sizes as discussed in the data section. Taken together, the short- and long-run results paint a consistent picture of a policy that durably reduced smoking prevalence in treated states.

V.B Intensive Margin Outcomes

As the new legislation increases awareness of the side effects of smoking, the distribution of daily cigarettes smoked could shift left—reflecting reduced consumption among continuing smokers—or right—reflecting the exit of light smokers from the pool. Which direction dominates in the intensive margin is an empirical question that I address next. To do so, I use data on self-reported smoking intensity in the last month. A challenge is that smokers tend to report factors of five when asked about their daily consumption, partly because cigarettes are sold in boxes of 10 or 20 as single-unit sales are illegal. To overcome this, I discretize consumption into bins of up to 5 cigarettes per day, between 5 and 10, between 10 and 20, and more than 20, and estimate equation (1) using an indicator for daily consumption in bin b as the dependent variable.¹³

The pre-policy coefficients for 2005, 2008, and 2009 are small and statistically indistinguishable from zero across all bins, supporting the parallel trends assumption for this outcome. The post-policy estimates reveal a rightward shift in the distribution. By 2013, the proportion of smokers consuming between zero and five cigarettes per

¹³Note that the shares of smokers in each bin add up to one and that the coefficients for each year add up to zero since they represent the changes in accumulated mass in the distribution of smoked cigarettes.

day decreases by 10.73 percentage points, while the proportion consuming more than 20 cigarettes per day increases by 2.69 percentage points.¹⁴ This indicates that the reduction in smoking prevalence documented in the previous section is partially driven by the exit of light smokers from the pool. This compositional shift has an important policy implication: the 2011 law may have reached a relatively receptive margin of the smoking population first, and further reductions in smoking prevalence will likely require complementary interventions targeted at heavier, more addicted smokers.

V.C Heterogeneous Effects

Do the effects of stricter restrictions on tobacco consumption vary across populations? In Figure 3 I address this question by analyzing the effects of the 2011 law on the probability of being a current smoker by age group and educational attainment.

The effects of the national law vary across age groups. The largest reductions in the probability of being a current smoker are concentrated among the youngest adults (18–25) and among adults aged 45–55, with both groups showing statistically significant effects. Adults aged 55 to 65, by contrast, did not respond to the increase in the cost of smoking induced by the national law. This pattern is consistent with the youngest adults responding primarily through never taking up smoking, while adults aged 45 to 55 respond through cessation of an existing habit. In 2013, among individuals aged 18 to 25, 66.5% were never smokers and 26.7% were current smokers; among those aged 45 to 55, only 46.9% were never smokers despite a comparable rate of current smokers (28.8%). Thus, while current smoking rates are similar across these two groups, the composition of non-smokers differs markedly—younger non-smokers are predominantly never smokers, while older non-smokers are predominantly former smokers, suggesting the two groups responded to the policy through different margins. One plausible explanation for the null effect among older adults is that smokers face greater difficulty quitting the longer they have been addicted to nicotine. Consistent with this hypothesis, intention-to-quit data reveal that only 60% of smokers aged 55 to 65 have ever intended to quit, compared to 67% among those aged 45 to 55, suggesting that discouragement accumulates with age.

Turning to education, Cutler and Lleras-Muney (2010) document a strong education gradient in smoking in the United States and United Kingdom, where college graduates are substantially less likely to smoke than those with lower educational

¹⁴This analysis uses data on consumption habits of smokers. I restrict the intensive margin analysis to the short-run sample, as the longer-run estimates are too imprecise to draw meaningful conclusions.

attainment. In my sample, pre-policy smoking rates confirm a similar pattern: 37% of elementary school graduates were current smokers, compared to 33% among high school graduates and 27% among college graduates. Despite these differences in baseline smoking rates, panel (b) of Figure 3 shows that the absolute reduction in the probability of being a current smoker is broadly similar across all three groups, with point estimates of comparable magnitude. In relative terms, however, the effects are somewhat larger for college graduates, for whom the reduction represents roughly 19% of their baseline smoking rate, compared to 14% and 12% for elementary and high school graduates respectively. This suggests that while the law reached smokers across the educational spectrum equally in absolute terms, more educated individuals experienced a proportionally larger reduction in smoking prevalence.

VI Effects of the 2011 National Law on Health Outcomes

In this section, I examine whether the reductions in smoking participation documented above translate into improved health outcomes. Though smoking can potentially damage nearly every part of the human body, I focus on outcomes strongly associated with smoking behavior and secondhand smoke exposure. Cigarette smoking is the leading cause of lung cancer and the primary driver of COPD, which includes emphysema and chronic bronchitis. Estimates from the U.S. Department of Health and Human Services indicate that smoking causes about 90% of all lung cancer deaths and 80% of all deaths from COPD.¹⁵ Alcaraz et al. (2016) estimate that in 2015 in Argentina, cigarette smoking caused 75% of COPD deaths and 82% of lung cancer deaths.

The outcomes of interest are defined as the ratio between the number of hospitalizations with diagnosis c in age group a and the total population aged a of state s in year t , multiplied by 1,000—that is, the prevalence rate of diagnosis c for age group a .¹⁶ I estimate the effects of the policy via the following equation, run over the sample of treated and comparison states for the period 2008-2014, by cause of discharge c for individuals aged 18 to 65:

$$y_a = \sum_{\substack{\tau=-3 \\ \tau \neq 0}}^3 \delta_\tau [Treat_s \cdot (Years\ After\ Treat = \tau)] + \Gamma' X_{st} + \alpha_a + \alpha_s + \alpha_t + \varepsilon_{st} \quad (2)$$

¹⁵See HHS, *The Health Consequences of Smoking—50 Years of Progress: A Report of the Surgeon General*.

¹⁶I compute prevalence rates by age group since population estimates for the period of analysis are not available by single ages.

where $Years\ after\ treat = \tau$ equals the difference between the calendar year and the year the national law was passed. The coefficients δ_τ can be interpreted as changes in the prevalence rate of diagnosis c when regulation was tightened. X_{st} is a vector of state-level control variables. Age-group fixed effects (α_a) ensure that δ_τ is identified by within age-group differences between lenient and strict states over time. Standard errors are robust to heteroskedasticity.¹⁷

Figure 4 presents the results. The pre-period coefficients for both COPD and lung cancer are small and statistically indistinguishable from zero, consistent with treated and comparison states being on similar trends prior to the policy change. Turning to the post-policy period, I find a gradual reduction in COPD hospitalization rates in lenient states following the tightening of regulation. The estimated coefficients display a negative gradient, with the effect growing in magnitude in each successive year after the law was passed. By 2014, the estimated reduction in COPD hospitalization rates is approximately 0.036 per 1,000 individuals, representing about 10.5% of the pre-policy mean in lenient states (0.3415 per 1,000; Table 1). The 2013 and 2014 coefficients are statistically significant at the 5% and 10% levels, respectively.

These results are consistent with the broader literature showing that many health conditions associated with smoking—such as heart disease, lung cancer, and overall premature mortality—develop over long periods of time, making it difficult to identify the appropriate lag between policy implementation and disease onset (DeCicca, Kenkel and Lovenheim, 2022). In the case of COPD, however, Eklund et al. (2012) show that smoking cessation and reduced exposure to cigarette smoke slow the excess decline in lung function, consistent with relatively faster health gains. Thus, the reduction in COPD hospitalization rates likely reflects the health benefits of smoking cessation and reduced exposure to secondhand smoke induced by the law. The null effects on lung cancer, by contrast, reflect the longer time horizon required to reverse the onset of this disease.

These reductions in hospitalization rates could be driven by fewer individuals being diagnosed with these diseases, by the same number of individuals presenting less severe symptoms and requiring fewer hospitalizations, or by a combination of both. As such, these results should be interpreted as suggestive evidence of an improvement in population health associated with stricter tobacco legislation. Unfortunately, the discharge data are not available beyond 2014, which precludes studying whether these

¹⁷I refrain from using clustered standard errors at the state level because the number of observations within each cluster is small.

effects persist or fade over a longer horizon.

VII Robustness Checks

Did state governments enforce the 2011 national law? The credibility of the estimated effects hinges on whether the national law was effectively enforced. To assess compliance, I exploit the fact that the national law banned smoking in bars and restaurants, which were unregulated in lenient states prior to 2011, while other venues such as workplaces, educational institutions, and hospitals were already regulated in both strict and lenient states. If the law was enforced, one would expect a larger reduction in environmental tobacco smoke exposure in bars and restaurants in lenient states relative to strict states, with little differential change in other venues. Figure A.2 presents estimates of the change in environmental tobacco smoke exposure across venues between 2009 and 2013.¹⁸ The results support this prediction: lenient states experienced a decline of approximately 23% in bars and restaurants relative to the 2009 mean—more than three times the reduction observed in other venues. This differential response is consistent with the law being enforced where it represented a genuine change in regulation.

Did the 2011 policy change prices of cigarettes? A potential concern is that the 2011 law may have induced changes in cigarette prices, which would confound the estimated effects on smoking behavior. To address this, I use data from the Ministry of Agriculture, which tracks cigarette sales at the national level and provides information on sales by price range paid by the consumer. I focus on three price ranges: the minimum, average, and maximum price paid in the cigarette market. These data are available at the price-range month level, allowing me to closely follow the evolution of prices around the date of the regulatory change. I use the price index constructed by Cavallo (2013) to construct real price series. Figure A.3 presents the evolution of real and nominal prices. There is no evidence of sharp changes in cigarette prices following the law. I pay particular attention to the minimum price, since smokers could substitute toward cheaper brands to offset any price increases in more expensive ones; the absence of a discontinuity in minimum prices further supports the conclusion that

¹⁸Environmental tobacco smoke (ETS) exposure questions were only asked in the 2009 and 2013 survey waves, limiting the analysis to a pre-post comparison. For each venue, I estimate $ETS_{ist} = \alpha + \delta(Treat_s \times Post_t) + \gamma Post_t + \alpha_s + X'_{ist}\beta + \varepsilon_{ist}$, where $Post_t$ is an indicator for 2013, α_s are state fixed effects, and X_{ist} is a vector of individual controls. Effect sizes are normalized by the 2009 venue-specific mean.

the policy did not operate through a price channel.

Leave-one-out exercise. A potential concern is that the results are driven by a single influential state. To address this, Figure A.4 presents leave-one-out estimates of the effect of the national law on the probability of being a never smoker and a current smoker. Each point corresponds to a regression run on a sample that excludes one state at a time; the plotted coefficient captures the effect of the law in 2013, the first post-policy survey year, estimated on the short-run sample. The estimates are remarkably stable across all leave-one-out samples: no single state drives the results, and all point estimates remain close to the full-sample estimate, shown in orange. This provides reassurance that the main findings are not sensitive to the inclusion of any particular state.

VIII Conclusion

Smoking bans and advertising regulations have played an increasingly important role in tobacco control policies over the past 20 years. These policies are particularly attractive in low- and middle-income countries, where tobacco use is concentrated among low-income households. This paper provides evidence of the effectiveness and potential limitations of these policies by studying the Argentinean case. Argentina's 2011 national law has two main components: (i) it implements indoor smoking bans and (ii) it regulates advertising of tobacco products and mandates the incorporation of graphic tobacco warnings. I exploit regional variation in the leniency of tobacco regulation before 2011 to identify the effects of this new regulation on smoking behavior and health outcomes. I find that the new regulations effectively curbed smoking initiation and smoking prevalence. Nonetheless, the heterogeneity analysis suggests that the policy reduced smoking proportionally more among the more educated, and thus may widen health disparities across education groups. The law's inclusion of graphic warnings is consistent with a growing body of evidence that package warnings can raise awareness and deter smoking.

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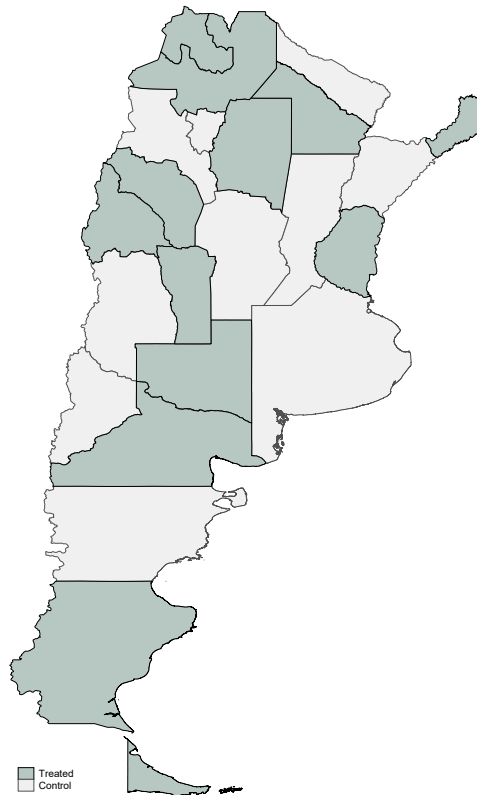
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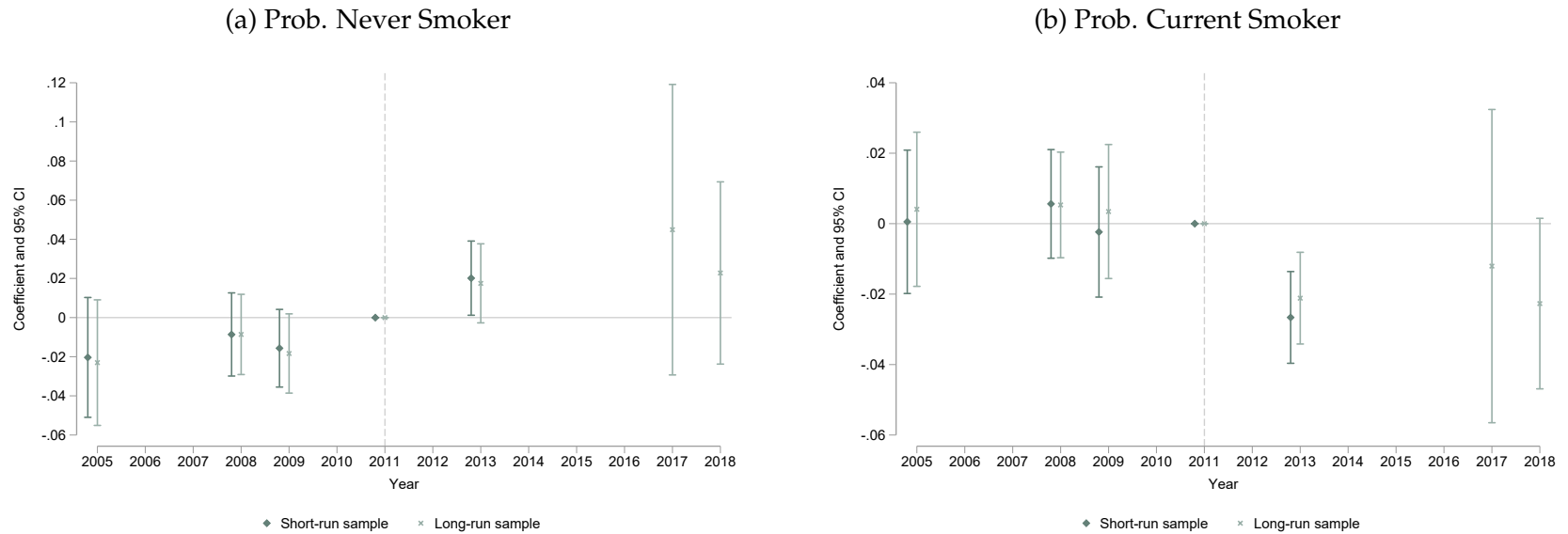
IX Figures

Figure 1: Treated vs Untreated States



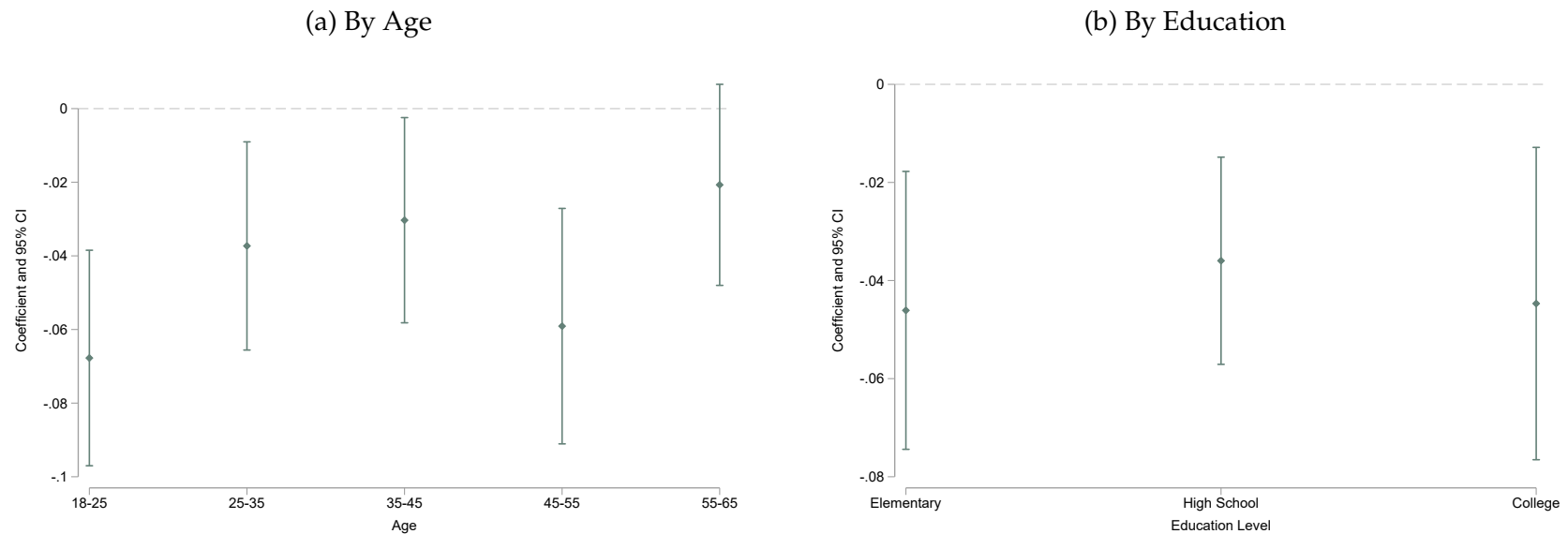
Notes: This figure presents the geographic distribution of treatment status across states. Dark shading indicates lenient states, which had weaker regulations prior to the 2011 national law and constitute the treated group. Light shading indicates strict states, which had stronger regulations prior to the 2011 national law and constitute the control group. Table [A1](#) provides details on each state's regulatory status and treatment classification.

Figure 2: Effects on Extensive Margin Outcomes



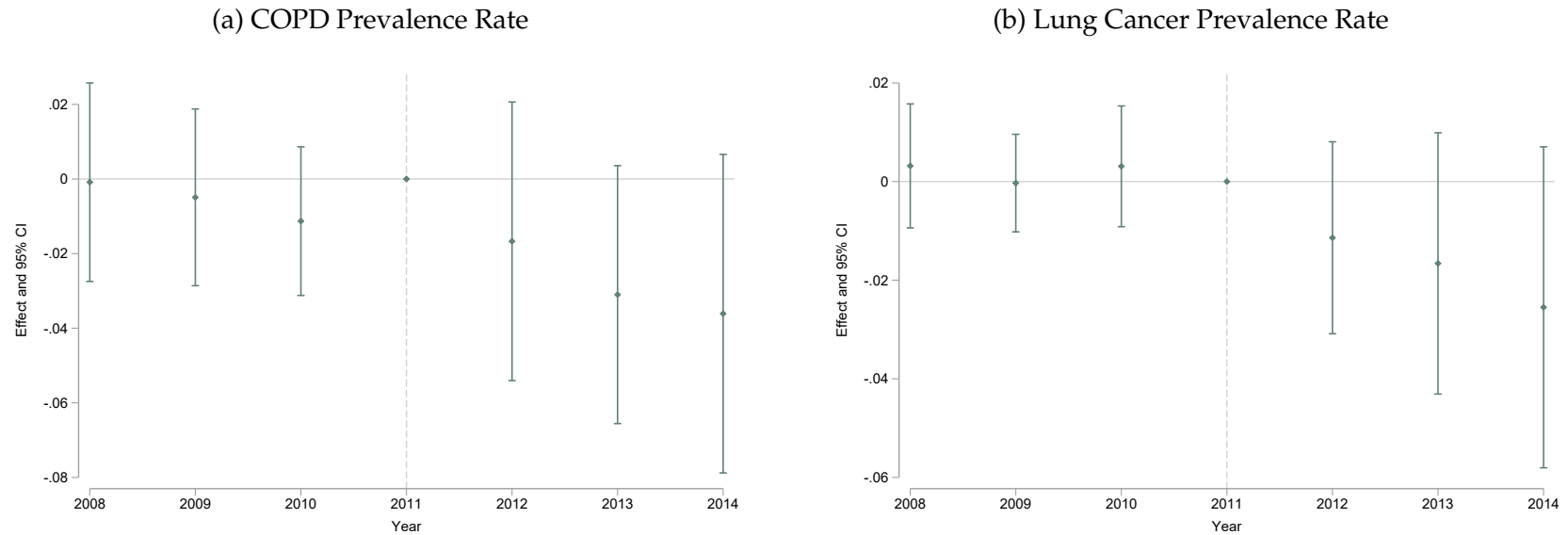
Notes: This figure presents point estimates and symmetric percentile-t confidence intervals for the causal effect of the national regulation on the probability of being a never smoker (left panel) and the probability of being a current smoker (right panel). The short-run sample uses data through 2013 only, while the long-run sample uses all years reported in the figure. The omitted year corresponds to 2011, the year the law was passed. Standard errors are block-bootstrapped at the state level with 300 replications.

Figure 3: Effects on Extensive Margin Outcomes by Age and Education – Current Smoker



Notes: This figure presents point estimates and symmetric percentile-t confidence intervals for the causal effect of the national regulation on the probability of being a current smoker by individual's age (panel a) and educational attainment (panel b). These estimates rely on the short-run sample, which uses data through 2013 only. The omitted year corresponds to 2011, the year the law was passed, and the figure presents the 2013 coefficients. Standard errors are block-bootstrapped at the state level with 300 replications.

Figure 4: Effects on Health Outcomes



Notes: This figure presents point estimates and symmetric percentile-t confidence intervals of the causal effect of the national regulation on the prevalence rate of COPD (left panel) and lung cancer (right panel). The omitted year corresponds to 2011, the year the law was passed. Standard errors are block-bootstrapped at the state level with 300 replications. The estimated reduction in COPD hospitalization rates is statistically significant at the 5% level in 2013 and at the 10% level in 2014; p-values are computed from the bootstrap distribution rather than from asymptotic approximations.

X Tables

Table 1: Summary Statistics, pre-policy period

Variables	All states (1)	Strict states (2)	Lenient states (3)	Equality of means (4)
Share of never smokers	0.5399	0.5364	0.544	0.7866
Share of current smokers	0.3092	0.3103	0.3079	0.8936
Mean cigarettes per day				
(0,5]	0.3894	0.4130	0.3615	0.2372
(5,10]	0.2650	0.2532	0.2791	0.0639
(10,20]	0.2765	0.2701	0.284	0.6171
20+	0.0690	0.0636	0.0754	0.1643
Prevalence rate (per 1,000)				
COPD	0.4205	0.4874	0.3415	0.0362
Lung cancer	0.1759	0.1898	0.1595	0.5172
Observations	62,183	30,846	31,337	
Number of states		11	13	

Notes: This table presents summary statistics for outcomes of interest for the pre-policy period. I pooled 2005, 2008, and 2009 for smoking habits outcomes and 2008 to 2010 for health outcomes. The last column presents the p-value for test of equality of means.

Table 2: Balance in Terms of Individual Characteristics in Strict versus Lenient States

Variables	All states (1)	Strict states (2)	Lenient states (3)	Equality of means (4)
Average age	37.94	38.23	37.70	0.0000
Male	0.4753	0.4727	0.4775	0.2330
Young (< 25 years old)	0.2333	0.2299	0.2362	0.0637
Married or cohabitant	0.5932	0.5907	0.5953	0.2405
Employed	0.6938	0.6918	0.6956	0.3005
Educational level				
Elementary school	0.0139	0.0133	0.0145	0.2099
High school	0.2871	0.2755	0.2968	0.0000
College	0.4296	0.4155	0.4416	0.0000
Income category				
First quintile	0.1517	0.1420	0.1599	0.0000
Second quintile	0.1588	0.1506	0.1656	0.0000
Third quintile	0.2465	0.2482	0.2450	0.3420
Fourth quintile	0.3439	0.3607	0.3297	0.0000
Fifth quintile	0.0991	0.0983	0.0998	0.5076
Observations	62,183	30,846	31,337	
Number of states		11	13	

Notes: This table presents summary statistics for demographic characteristics for the pre-policy period. I pooled 2005, 2008, and 2009. Demographic characteristics other than age indicate the proportion of individuals with a given attribute. The last column presents the p -value for test of equality of means.

Table 3: Determinants of Strong Regulation prior to 2011

Dependent variable:	Indicator for Enacting a Strong Regulation	
	Coefficient	Marginal Effect
Peronist party (=1)	0.390 (0.266)	0.133 (0.090)
State is tobacco producer (=1)	0.598 (0.372)	0.204 (0.128)
Ln employment	-0.172 (0.516)	-0.058 (0.177)
Unemployment rate	4.769 (5.197)	1.626 (1.783)
Proportion of smokers	-0.327 (3.060)	-0.112 (1.044)
Prevalence of COPD	1.156*** (0.378)	0.394*** (0.130)
Share population 0 - 14 years old	0.066 (16.370)	0.022 (5.584)
Share population 15 - 24 years old	48.93*** (17.580)	16.687*** (5.832)
Share population 25 - 44 years old	20.400 (16.510)	6.958 (5.682)
Share population 45 - 64 years old	44.210 (30.110)	15.077 (10.360)
Ln population	0.542 (0.529)	0.185 (0.183)
Observations	24	24
Pseudo R^2	0.2196	

Notes: This table presents results from a probit model estimating the determinants of enacting strict tobacco regulation prior to 2011. The estimated equation is: $Strict_s = \Phi(\alpha + \beta_1 Peronist_s + \beta_2 Producer_s + \beta_3 X_s + \varepsilon_s)$ where $Strict_s$ is an indicator equal to 1 if state s had strict pre-existing regulation, $Peronist_s$ indicates whether the state government was affiliated with the Peronist party, $Producer_s$ indicates whether the state is a tobacco producer, and X_s is a vector of state-level demographic and economic characteristics. Column (1) presents estimated coefficients and column (2) presents marginal effects computed at the means. The Peronist party was responsible for enacting the 2011 national law; the coefficient on the Peronist party indicator therefore tests whether states governed by the Peronist party were more likely to regulate tobacco at the state level relative to opposition-governed states. States classified as tobacco producers are Jujuy (36% of national production), Misiones (29%), Salta (25%), Tucuman (7%), Catamarca (1%), Corrientes (1%), and Chaco (1%). Standard errors are robust to heteroskedasticity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects on Intensive Margin Outcomes

Cigarettes smoked	(0, 5] (1)	(5, 10] (2)	(10, 20] (3)	(20+] (4)
2005	0.031 (0.0198)	0.0057 (0.0119)	-0.0326* (0.0143)	-0.0041 (0.0085)
2008	-0.0033 (0.0133)	0.0133 (0.0122)	-0.0095 (0.0148)	-0.0005 (0.0064)
2009	-0.0135 (0.0178)	0.0262 (0.0195)	0.003 (0.025)	-0.0157 (0.0125)
2013	-0.1073*** (0.0174)	0.0527*** (0.0151)	0.0277 (0.0183)	0.0269*** (0.0096)
Mean of dep variable	0.262	0.2956	0.3426	0.0997
Observations	40,945	40,945	40,945	40,945

Notes: The omitted category corresponds to 2011, the year the federal law was passed. Treatment is a dichotomous variable, $Treat_s$, that equals one for lenient states. All regressions include state fixed effects, year fixed effects, state time varying controls and individual level controls. State $\times$ time controls include total private employment and total population. Individual level controls are: age, gender, educational attainment, employment status and income category of the household. Standard errors are block-bootstrapped at the state-level with 300 replications.

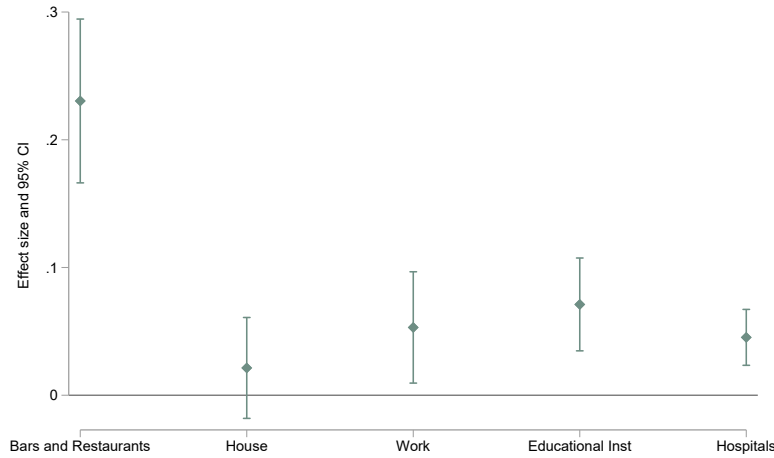
A Additional Figures and Tables

Figure A.1: Examples of law-regulated packaging.



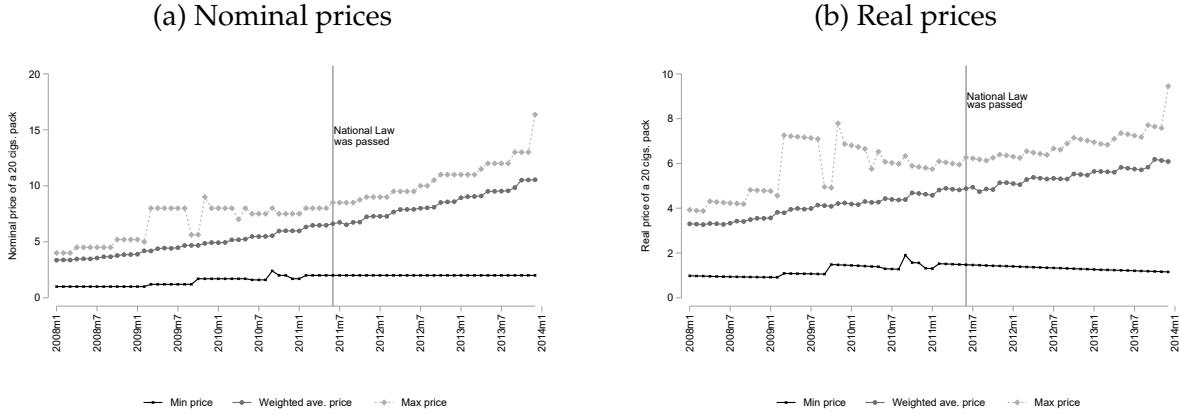
Notes: This figure provides an example of the graphic warnings printed on cigarette boxes. Big font messages: (i) smoking reduces years of life, (ii) smoking causes cancer, (iii) smoking might cause leg amputation, (iv) pregnant women who smoke harm her child (v) smoking causes death by suffocation. Small font messages: (i) tobacco drives half of smoker's deaths, (ii) every cigarette poisons you, (iii) smoking causes gangrene, (iv) tobacco is harmful for fetal development, (v) every cigarette damages your respiratory capacity.

Figure A.2: Change in Exposure to Environmental Tobacco Smoke



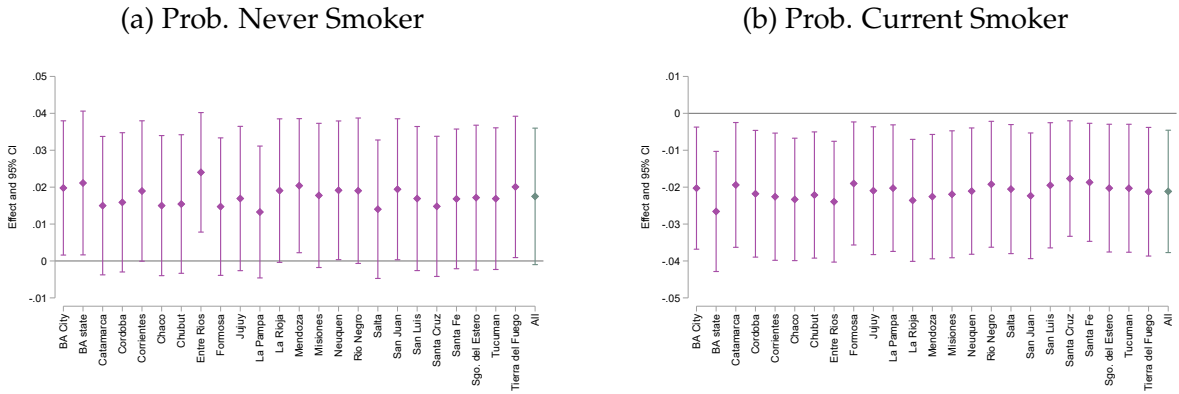
Notes: This figure presents estimates of the change in exposure to environmental tobacco smoke across venues between 2009 and 2013, the only two survey years in which ETS exposure questions were asked. For each venue, I estimate $ETS_{ist} = \alpha + \delta(Treat_s \times Post_t) + \gamma Post_t + \alpha_s + X'_{ist}\beta + \varepsilon_{ist}$, where $Post_t$ is an indicator for 2013, α_s are state fixed effects, and X_{ist} is a vector of individual controls. The coefficient δ captures the differential reduction in ETS exposure in lenient states relative to strict states. Effect sizes are normalized by the 2009 venue-specific mean. Confidence intervals are computed using robust standard errors.

Figure A.3: Evolution of Nominal and Real Cigarettes Prices



Notes: This figure presents time series of real and nominal prices paid by consumers split into three price ranges: the minimum price, the average price, and the highest price. I use the price index constructed by Cavallo (2013) to deflate prices. Data are at the price-monthly level and come from price series constructed by the Ministry of Agriculture.

Figure A.4: Sensitivity of Extensive Margin Outcomes to State Exclusion



Notes: This figure presents leave-one-out estimates of the effect of the national law on the probability of being a never smoker (panel a) and a current smoker (panel b). Each point estimate corresponds to a regression run on a sample that excludes one state at a time; the rightmost point in orange corresponds to the estimate using the full sample. The plotted coefficient captures the effect of the national law in 2013, the first post-policy survey year. All regressions are estimated on the short-run sample, which uses data through 2013, and include individual controls, state fixed effects, and year fixed effects, following the specification in equation 1. Confidence intervals are computed using cluster-robust standard errors at the state level.

Table A1: State level regulation

State	Laws codes and dates (1)	Short description (2)	Treatment (3)
Buenos Aires	Law 13,894 (2008)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation, educational institutions, healthcare facilities, and restaurants.	0
Catamarca	Law 5,223 (2007)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in educational institutions, healthcare facilities, and restaurants.	0
Chaco	Law 5,544 (2005)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation.	1
Chubut	Law 3,775 (1993)	Restricts sales to underage individuals. Limits smoking in public transportation and healthcare facilities.	1
Ciudad de Buenos Aires	Law 1,799 (2005) and law 3,718 (2010)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation, educational institutions, healthcare facilities, and restaurants.	0
Cordoba	Law 9,113 (2003)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation, and healthcare facilities.	0
Corrientes	Law 5,537 (2004) and law 5,793 (2007)	Restricts sales to underage individuals. Limits smoking in public transportation, educational institutions, and healthcare facilities.	0
Entre Rios	Law 9,862 (2008)	Limits smoking in educational institutions, healthcare facilities, and restaurants.	1
Formosa	Law 1,103 (1994)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation and healthcare facilities.	0
La Pampa	Law 2,563 (2010)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation, educational institutions, healthcare facilities, and restaurants.	0

(continue)

Table A1: State level regulation (*continued*)

State	Laws codes and dates (1)	Short description (2)	Treatment (3)
La Rioja	Law 7,525 (2003) and Law 8,870 (2010)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation.	1
Mendoza	Law 7,790	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation, healthcare facilities, and restaurants.	0
Misiones	Law 1,670 (2009)	Limits smoking in public transportation, educational institutions, healthcare facilities, and restaurants.	1
Neuquén	Law 2,738 (2010) and law 2,572 (2008)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation, healthcare facilities, and restaurants.	0
Río Negro	Law 3,986 (2005)	Regulates advertising. Limits smoking in public transportation, and healthcare facilities.	1
Salta	Law 7,631 (2010)	Limits smoking in public transportation.	1
San Juan	Law 7,595 (2005)	Limits smoking in public transportation.	1
San Luis	Law 5,451 (2004)	Limits smoking in public transportation, educational institutions, and healthcare facilities.	1
Santa Cruz	-	No state law before 2013 – Law 3,329	1
Santa Fe	Law 12,432 (2005)	Regulates advertising. Restricts sales to underage individuals. Limits smoking in public transportation, educational institutions, and healthcare facilities.	0
Santiago del Estero	Law 6,962 (2009)	Limits smoking in public transportation, educational institutions, and healthcare facilities.	1
Tierra del Fuego	Law 175 (1994)	Limits smoking in public transportation.	1
Tucumán	Law 7,575 (2005)	Regulates advertising. Limits smoking in public transportation, educational institutions, and healthcare facilities.	0

Notes: Column (1) reports the law code and year of passage. Column (3) indicates treatment status, where treated states (indicated with a 1) had weaker pre-existing regulation and control states (indicated with a 0) had stricter regulation before the 2011 national law. Santa Cruz had no state-level legislation prior to 2013 and is therefore assigned to the treatment group.

Table A2: Evolution of Sales: Administrative Records and Survey Data

Year	Sales data		Survey data	
	Consumption	Percent change	Consumption	Percent change
2008	181.06		106.75	
2009 ^(a)	177.64	-1.89%	103.42	-3.12%
2010	174.86			
2011 ^(b)	182.65	1.40%	105.84	1.16%
2012	178.36			
2013 ^(b)	174.07	-2.38%	99.39	-3.09%

Notes: Consumption measures are expressed in millions of 20-cigarette packages per month. Sales data are collected by the Ministry of Agriculture. Consumption from survey data is computed based on average reported consumption per day. Survey data are only available for 2008, 2009, 2011, and 2013. Percent changes are: (a) the percent change relative to the previous calendar year; (b) the annualized (compound) percent change relative to the $n - 2$ year, computed as $(\text{Consumption}_n / \text{Consumption}_{n-2})^{1/2} - 1$.

B Legislation Data Collection

Argentina has 23 states and one federal district.¹⁹ Each state has the autonomy to dictate laws in their territory as long as these laws are not contrary to federal law. Broadly speaking, taxes on tobacco-related products are set at the national level, but non-price policies had been put in place at the regional level before law 26,687 was passed. I coded a total of 47 regional laws and two national laws across three main categories: tobacco advertising, sales, and consumption. Each category is further divided into sub-categories, following [González-Rozada \(2006\)](#) report on the status of tobacco legislation in Argentina.

The advertising category includes the following sub-categories: publicity, advertising on radio, advertising on television, advertising directed at specific audiences, regulation of advertising content, event sponsoring, brand stretching, inclusion and size of TGW, and inclusion of contact information for anti-smoking public services. The sales category includes bans on sales: to individuals under 18, at elementary schools, at high schools, at educational institutions in general, at hospitals or health institutions, in government buildings, in public transportation, and sales by the unit. The consumption category includes the following sub-categories: government buildings, workplaces, health institutions, elementary schools, high schools, universities, public transportation, restaurants, bars, and entertainment centers.

Table [A1](#) presents the full list of state laws collected, including law codes, dates, and a short description of their main provisions. A state is coded as a control state if its pre-existing regulation covered at least three of the four consumption environments (public transportation, educational institutions, health institutions, and restaurants and bars), regardless of whether it also regulated advertising or sales to minors. States that did not meet this threshold are coded as treatment states.

¹⁹I will refer to this administrative unit as another state since the distinction between state and federal district is not relevant for this paper.